

# Agentic AI for Chemical Information Extraction from Scientific Publications

Kasper Marian Roland Huysentruyt

Promotors: prof. dr. ir. Kevin M. Van Geem, prof. dr. ir. Christian V. Stevens  
Counsellor: dr. ir. Maarten Dobbelaere

**Abstract:** The chemical literature grows faster than it can be read. The CAS REGISTRY now exceeds 279 million substances, yet almost all of this chemistry stays trapped in figures and PDFs that machines cannot parse. The structured reaction records that retrosynthesis, yield-prediction and condition-prediction models depend on are still built by hand, a process that is slow, costly and prone to transcription error. Recent vision-language models, placed inside tool-using agentic loops, could automate this work, but whether extraction quality is set by the model itself or by the orchestration around it has not been clear. Here we build and benchmark four extraction pipelines on sixteen reaction figures drawn from five chemistry journals, ranging from a large multi-agent system to a single model equipped with two chemistry tools. We find that the model, not the framework, decides the outcome. Holding prompt, tools and output schema fixed, swapping only the underlying model produced a 0.61-point spread in product-level accuracy (intersection-over-union), far wider than any change caused by altering the orchestration. The simplest single-agent design surpassed every multi-step system, with Claude Opus 4.7 giving the best overall extraction quality and Gemini 3 Flash Preview matching it at roughly one-tenth the cost. These results suggest that progress in automated reaction extraction will come mainly from stronger vision encoders rather than from more complex agent architectures.

**Keywords:** Chemical reaction extraction, vision-language models, agentic AI, multi-agent systems, structured data extraction, cheminformatics, LLM tool use, benchmark evaluation.

## I. INTRODUCTION

The chemical literature is expanding at an unprecedented rate. The CAS REGISTRY exceeded 279 million unique substances in early 2025, up from 105 million a decade earlier [1], and global publication output doubles roughly every seventeen years [2]. Yet almost all of this chemistry remains locked inside PDFs whose text and figures are not intrinsically machine readable [3]. Trusted reaction databases such as Reaxys and SciFinder are still curated by hand, which is slow, expensive, and prone to transcription error [4]. As a result, only a small fraction of the published reaction record is available in a structured form that downstream models can consume (Figure 1), and a substantial share of the underlying primary literature additionally sits behind paywalls that further constrain reuse. The goal of this thesis is to evaluate whether modern agentic AI, in which a vision-language model (VLM) is wrapped in a tool-using decision loop, can close the gap between the published reaction record and the structured tables that retrosynthesis, condition prediction, yield regression models and reaction-class analysis tools such as Rxn-INSIGHT [5] require [6, 7].

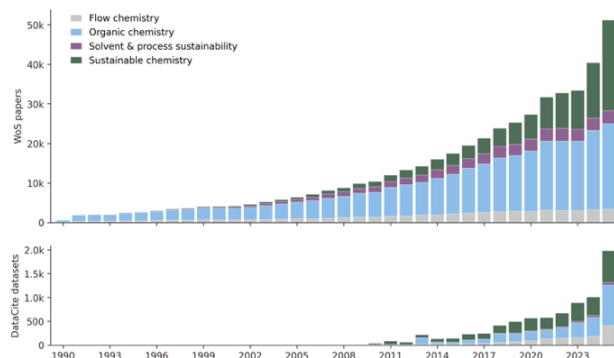


Figure 1: Number of research papers (top) versus structured datasets (bottom) deposited per year for four chemistry sub-fields. Even after two decades of growth in open data infrastructure, structured records lag publications by several orders of magnitude

Earlier generations of reaction-extraction pipelines relied on dictionary lookups, hand-written grammars and small machine-learning taggers trained on manually annotated corpora; tools such as ChemicalTagger, ChemDataExtractor and OSCAR exemplify this generation [4, 8, 9]. They achieve high accuracy on narrowly scoped inputs but scale poorly across the heterogeneous formats of contemporary chemistry papers. The advent of large language models (LLMs) and VLMs has shifted the landscape: a single multimodal model can now read a reaction figure end to end without task-specific fine tuning. Agentic frameworks extend this further by giving the model a decision loop, structured output schemas, and external tools (RDKit, optical chemical structure recognition (OCSR), and name-to-structure resolvers), so that symbolic chemistry primitives remain available alongside the model's generative reasoning [7]. The recently published ChemEAGLE pipeline [10] is representative of this shift and reports state-of-the-art numbers on its own benchmark, but its design space (number of agents, choice of backbone model, division of labour between deterministic tools and the VLM) has barely been mapped.

This thesis takes that mapping as its central research question: on a shared benchmark of sixteen chemistry figures, how much of the extraction quality is decided by the multimodal model itself, and how much by the orchestration that surrounds it? To answer this, four extraction pipelines were built and evaluated under identical conditions: (i) the published ChemEAGLE pipeline reinstalled locally as the reference baseline; (ii) an original multi-agent framework that integrates a deterministic R-group decomposition layer absent from ChemEAGLE; (iii) ChemEAGLE Hybrid, a variant designed independently; and (iv) a single-agent VLM pipeline instantiated behind nine multimodal models (Claude Opus 4.6 and 4.7, GPT-5.4 and 5.5, three Gemini variants, xAI Grok 4.3,

and the open-weights Qwen3.6-27B). Every system writes to the same Pydantic schema, is scored against the same ground truth, and is benchmarked on quality, cost, wall time and per-system failure modes.

## II. DATA AND METHODS

### A. Benchmark

All systems are evaluated on a curated 16-image subset of the GT3 ground-truth corpus published alongside ChemEAGLE [10]. The images cover reaction-scope tables, condition-survey panels and picture-style schemes, drawn from five journals. Each figure has been cropped from its journal page so no surrounding running text is available, which isolates the visual extraction task from out-of-figure context. The ground-truth schema is a flat list of reactions, each carrying reactant SMILES, product SMILES, and a conditions list whose role field is drawn from a closed enumeration covering reagent, solvent, catalyst, temperature, time, yield, atmosphere, loading and pressure. R-group placeholders are encoded as atom-mapped wildcards ([1\*], [2\*]). Sixteen figures are too few for fine-grained ranking, but sufficient for the qualitative cross-system comparison this thesis pursues, and small enough to keep the inference cost of a nine-model sweep within budget.

### B. Evaluation Methodology

Every system is scored against the same ground truth with the same evaluation harness. SMILES are canonicalised through RDKit before comparison; chirality is stripped because some ground-truth files use abbreviated stereo notation. Per-image reactant and product intersection over union (IoU) are micro-averaged Jaccard indices on the corresponding SMILES sets. Reaction-level matches are then computed under two

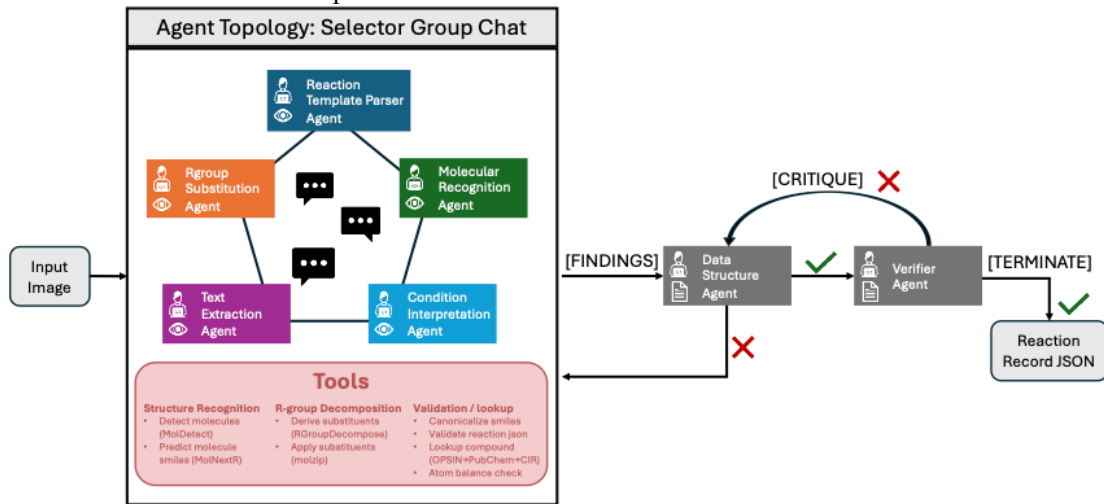
bucketing, SMILES-based name unification) and roughly doubles the strict scores, the gap interpreted as formatting noise rather than chemistry error.

### C. ChemEAGLE Reference Baseline

ChemEAGLE was reinstalled locally as the reference baseline. It coordinates a Planner with six specialist agents (template parser, molecular recognition, R-group substitution, condition interpretation, text extraction, data structuring) under two observer modules, and integrates Microsoft Florence-2 for figure detection, OCSR via MolNexTR [11], and OPSIN/PubChem/CIR for name-to-structure resolution [12-14]. Despite the published 0.78 soft-F1 on its own benchmark, the local rerun reached only 0.29 soft-F1 and crashed on four of the sixteen images inside the MolNexTR canonicalization step. The reproduction gap motivated two parallel research lines: an alternative multi-agent orchestration over the same chemistry toolchain, and a single-agent baseline that strips orchestration away entirely.

### D. Original Multi-Agent Framework

The original multi-agent framework wires seven agents into a Selector Group Chat [15] team in which five specialist agents emit plain-text findings cards on a shared image, the Data Structure agent assembles a single Pydantic-valid JSON, and a Verifier critiques and re-triggers up to two retry cycles (Figure 2). The development progressed through ten numbered phases. Three changes carried most of the headline gains: (a) a Pydantic schema with a retry loop lifted schema validity from below 0.50 to 1.00; (b) replacing the VLM’s ad-hoc molecule reading with a per-bounding-box MolDetect [16] and MolNexTR routine (Phase 8) doubled product IoU; and (c) writing a deterministic RDKit [17] RGroupDecompose-and-molzip chain (Phase 9)



**Figure 2: Topology of the original multi-agent framework. Five specialist agents (top row) emit findings cards on the shared image; the Data Structure agent is the only JSON producer; the Verifier critiques and re-prompts up to two retries before forced termination.**

criteria: a soft match requires the reactant and product sets to coincide after canonicalization; a hard match additionally requires the condition set to match. A bipartite (Hopcroft–Karp) matcher aligns predicted and ground-truth reactions to neutralise ordering, and precision, recall and F1 are reported under both criteria together with a partial-match F1 based on a Jaccard-0.5 threshold and a constitution-F1 with stereochemistry stripped. Conditions are scored separately under strict and lenient matchers; the lenient matcher absorbs cosmetic differences (whitespace, parenthesisation, role

raised reactant IoU on the canonical R-group case from 0 to 1.0. The combined posture, constrain the output, delegate perception to specialist deep-learning models, and reserve symbolic manipulation for deterministic code, is the design principle that the original multi-agent framework operationalises.

### E. Single-Agent Pipeline

The single-agent pipeline strips the orchestration to its simplest form: one multimodal model with two tools (an

RDKit-backed SMILES validator and a Pydantic-validated extraction terminator) and one system prompt that mirrors Anthropic’s prompt-engineering recommendations for Opus 4.7 [18]. The agent loop is implemented once per provider and shares the same schema, the same tools and the same prompt across nine models: Claude Opus 4.6 and 4.7 via the Anthropic SDK [18, 19]; GPT-5.4 and 5.5 via Microsoft Azure AI Foundry [20, 21]; Gemini 3.1 Pro Preview, Gemini 3 Flash Preview and Gemini 3.5 Flash via the Google Generative AI SDK [22-24]; xAI Grok 4.3 via the xAI API [25]; and the open-weights Qwen3.6-27B served through OpenRouter [26]. The five provider adapters differ only in their wire-format translation layer, which means any quality difference across models is attributable to the model itself rather than to changes in agent design. Substrate-scope figures (Figure 3) are particularly demanding because most variants must be reconstructed by structural reasoning from the products rather than read directly off the page; they form the discriminating case in the cross-system comparison.

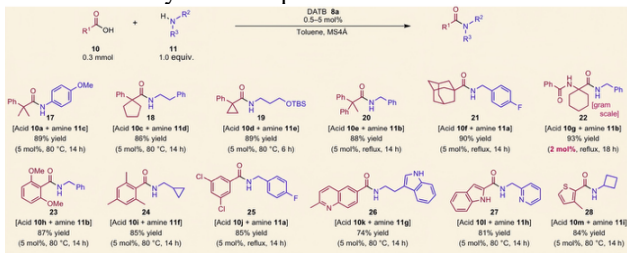


Figure 3: Substrate-scope figure (NC\_2017, page 4). Only the generic template (top) and concrete products (bottom) are drawn; variant reactants must be reconstructed by the deterministic RDKit R-group chain (Phase 9).

### III. RESULTS AND DISCUSSION

#### A. Phase Progression of the Multi-Agent Framework

Inside the multi-agent framework, the phase-by-phase progression on the canonical R-group case (Figure 4) is the cleanest illustration of where extraction quality is decided. Product IoU stabilises at 1.0 once MolNexTR is invoked once per detected bounding box (Phase 8); reactant IoU on the same image only reaches 1.0 after the deterministic RDKit chain replaces VLM reasoning (Phase 9). Both steps come from delegating sub-tasks to specialised models or deterministic code rather than to the VLM.

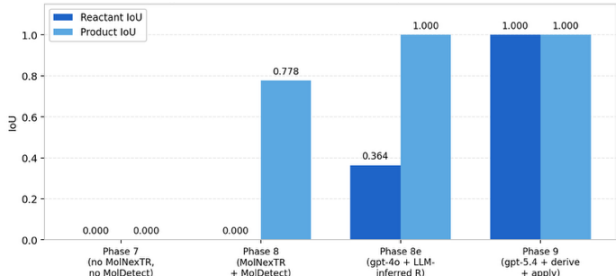


Figure 4: Phase progression on a reaction image with 8 reactions. Product IoU stabilises at 1.0 with per-bbox MolNexTR (Phase 8); reactant IoU reaches 1.0 only after the deterministic RDKit chain (Phase 9).

#### B. Cross-System Comparison

On the full sixteen-image benchmark, all twelve evaluated systems land in a heatmap that puts the strong single-agent variants clearly above every multi-step baseline on the structural metrics (Figure 5). Opus 4.7 achieves the highest

Partial F1 of 0.85, surpassing all other pipelines. However, Gemini 3 Flash Preview attains product IoU of 0.79 and Claude Opus 4.7 attains 0.76, both about 0.21-0.24 above the strongest multi-step pipeline (the original multi-agent framework at 0.55). Within the single-agent family, however, the spread is dramatic: GPT-5.4 collapses to product IoU of 0.18, the lowest in the table. This is due to consistent vision-encoder misreads of specific atom-count and connectivity motifs. Identical agent infrastructure, holding prompt, tools and schema constant, therefore produces a 0.61-point absolute spread on product IoU when only the multimodal model is varied. This spread is substantially larger than any orchestration-driven gap observed in the comparison, and it is what motivates the central claim of this thesis: on this benchmark, the multimodal model itself is the dominant determinant of extraction quality.

|                          | Product IoU | Reactant IoU | Soft F1 | Partial F1 ( $\geq 0.5$ ) | Cond. recall (len=) |
|--------------------------|-------------|--------------|---------|---------------------------|---------------------|
| multi_agent              | 55%         | 39%          | 22%     | 37%                       | 77%                 |
| ChemEagle                | 39%         | 45%          | 29%     | 39%                       | 55%                 |
| ChemEagle_Hybrid         | 45%         | 32%          | 18%     | 20%                       | 32%                 |
| Opus 4.7 *               | 76%         | 68%          | 60%     | 85%                       | 77%                 |
| GPT-5.4 *                | 18%         | 24%          | 9%      | 27%                       | 71%                 |
| Gemini 3.1 Pro Preview * | 57%         | 59%          | 40%     | 83%                       | 85%                 |
| Gemini 3 Flash Preview * | 79%         | 57%          | 46%     | 79%                       | 81%                 |
| Grok 4.3 *               | 64%         | 53%          | 42%     | 78%                       | 74%                 |
| Opus 4.6 *               | 55%         | 55%          | 41%     | 64%                       | 78%                 |
| GPT-5.5 *                | 62%         | 56%          | 43%     | 72%                       | 76%                 |
| Qwen3.6-27B *            | 40%         | 54%          | 40%     | 66%                       | 80%                 |
| Gemini 3.5 Flash *       | 62%         | 49%          | 38%     | 80%                       | 82%                 |

Figure 5: Per-system heatmap of the 12 evaluated extraction systems on the 16-image benchmark (colour scaled per column, rows marked with \* are single-agent runs). Strong single-agent variants (Opus 4.7, GPT-5.5, Gemini family, Grok 4.3) outperform every multi-step baseline on the structural metrics.

#### C. Cost-Quality Trade-off

The cost-quality picture (Figure 6) makes the practical case. Claude Opus 4.7 leads on partial-F1 at 0.85 but at \$0.22 per image; Gemini 3 Flash Preview reaches partial-F1 0.79 at \$0.018 per image, the cheapest competitive option in the comparison; Gemini 3.5 Flash sits between them at \$0.055 per image with a partial-F1 of 0.80; and the open-weights Qwen3.6-27B served via OpenRouter sits near the fitted line at \$0.085 per image with a partial-F1 of 0.66. The original multi-agent framework is clearly off the Pareto front: the highest cost in the table for a partial-F1 below 0.40. Wall time and token spend tell the same story (Figure 7 and Figure 8): Opus 4.7 is simultaneously the fastest single-agent variant (41s per image) and the highest scoring, with no monotonic trend linking tokens spent to quality ( $R^2$  of the log-linear fit is essentially zero,  $R^2 = 0.00$ ). The choice of model and pipeline shape dominates raw resource budget on this task.

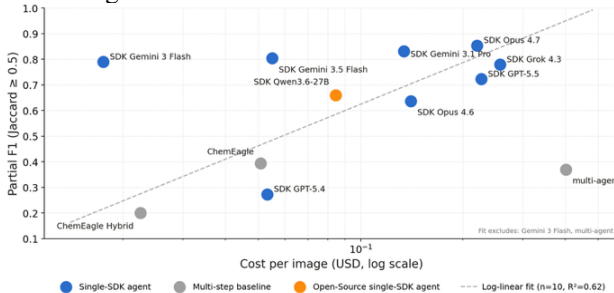
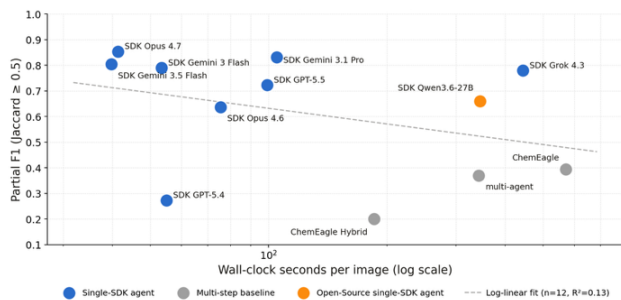
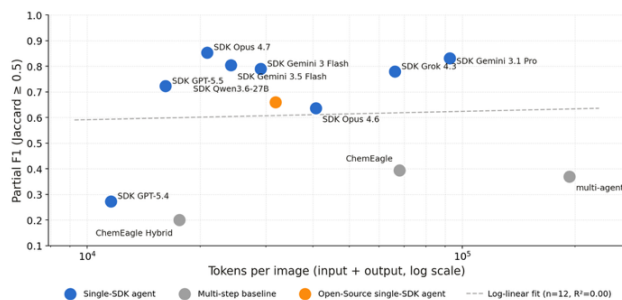


Figure 6: Cost vs. partial-F1 (log x). Claude Opus 4.7 leads on partial-F1, but at a high cost compared to Gemini 3 Flash. Gemini 3 Flash Preview and the multi-agent framework are excluded as outliers from the trend ( $R^2=0.62$ ).





**Figure 7: Wall-clock seconds vs. partial-F1 (log x). Opus 4.7 leads on both quality and speed (41 s/image); the trend is slightly negative ( $R^2 = 0.13$ ).**



**Figure 8: Tokens per image vs. partial-F1 (log x). Multi-agent consumes the most tokens per image. The log-linear fit is essentially flat ( $R^2 = 0.00$ ); token spend is not predictive of quality.**

#### D. Failure Modes

Per-image inspection of the low-scoring cases attributes most multi-agent failures to two structural weaknesses. First, MolDetect was trained on chemistry-figure crops and treats dense text fragments as molecule boxes when fed page-cropped scans; MolNexTR then fabricates RDKit-valid but chemically wrong SMILES, and the verifier loop never fires. Second, on dense substrate-scope grids the agent sometimes deposits intermediate R-group fragments into the output JSON without re-invoking the apply substituents step, a non-deterministic adherence failure that no prompt tightening resolved. On the single-agent side, the four-image Opus 4.6 versus Opus 4.7 case study traces the headline gap to perception events at the input stage (an isopropyl group read as methyl, a  $\text{CH}_2$  linker read as a carbonyl, a benzothiophene confused with a thiophene). When vision fails, Opus 4.6 falls back to chemically plausible priors from training data rather than to low-confidence output; Opus 4.7 with its 3.75-megapixel image cap reads the page more literally. The implication is that improvements in the vision encoder, not in the orchestration around it, are what minimises the remaining quality gap on this benchmark.

### IV. CONCLUSION AND FUTURE WORK

#### A. Conclusion

The multimodal model itself is the dominant variable in agentic chemical reaction extraction: identical agent infrastructure produces a 0.61-point spread in product IoU when only the model is varied, much larger than the spread produced by varying the orchestration at fixed model. The framework matters when the model is weak; the model matters more than the framework when the model is strong. Among the evaluated systems, Claude Opus 4.7 achieves the highest quality at the highest cost, Gemini 3 Flash Preview is the

cheapest competitive option, and Gemini 3.5 Flash sits between them with full schema coverage at a quarter of Opus 4.7's per-image cost. Every multi-step pipeline, including the original multi-agent framework, ChemEAGLE and ChemEAGLE Hybrid, is outperformed in partial-F1 by the best single-agent configurations at comparable or lower cost. The multi-agent value proposition is therefore no longer highest partial-F1 but agent-level auditability and guaranteed SMILES validity through an enforced RDKit round-trip, which matter for downstream regulatory documentation and integration with structural verifiers.

#### B. Future Work

Three follow-on directions are flagged. First, the benchmark itself should be extended with multi-annotator ground truth and inter-annotator agreement scores, both to break the dependence on the sample and to support fine-grained model ranking. Second, the proprietary models that currently dominate the cost-quality Pareto are accessed through provider APIs and have a finite useful lifetime; distilling the Opus 4.7 single-agent behaviour into a smaller open-weights student (Qwen3.6-27B as the natural first target) would make the pipeline durable against vintage shifts and enable on-premises deployment for groups whose data cannot leave the lab. Third, the cost-quality Pareto front motivates hybrid routing in which the agent decides per image which model to query and cascades from a fast model to a more expensive one only when the fast model's confidence is low. Beyond the benchmark, the cost numbers reported here (\$0.02 to \$0.22 per image, under one minute per image with Gemini 3 Flash Preview) are low enough that the extractor can live inside interactive chemist workflows and inside the literature-conditioning step of the emerging autonomous-lab platforms that closed-loop chemistry.

### V. REFERENCING

- (CAS), C.A.S. *Annual Report 2015*. 2016. 29 oct 2025].
- Larsen, P. and M. Von Ins, *The rate of growth in scientific publication and the decline in coverage provided by Science Citation Index*. *Scientometrics*, 2010. **84**(3): p. 575–603.
- Rajan, K., et al., *DECIMER-Segmentation: Automated extraction of chemical structure depictions from scientific literature*. *Journal of Cheminformatics*, 2021. **13**(1): p. 20.
- Hawizy, L., et al., *ChemicalTagger: A tool for semantic text-mining in chemistry*. *Journal of Cheminformatics*, 2011. **3**(1): p. 17.
- Dobbelaere, M.R., et al., *Rxn-INSIGHT: fast chemical reaction analysis using bond-electron matrices*. *Journal of Cheminformatics*, 2024. **16**(1): p. 37.
- Schilling-Wilhelmi, M., et al., *From text to insight: large language models for chemical data extraction*. *Chemical Society Reviews*, 2025. **54**(3): p. 1125–1150.
- Schwaller, P., et al., *Machine intelligence for chemical reaction space*. *WIREs Computational Molecular Science*, 2022. **12**(5): p. e1604.
- Mavračić, J., et al., *ChemDataExtractor 2.0: Autopopulated Ontologies for Materials Science*. *Journal of Chemical Information and Modeling*, 2021. **61**(9): p. 4280–4289.
- Jessop, D.M., et al., *OSCAR4: a flexible architecture for chemical text-mining*. *Journal of Cheminformatics*, 2011. **3**(1): p. 41.
- Chen, Y. and C.Y.F. Group, *ChemEagle: A Multi-Agent Pipeline for Structured Reaction Extraction from Chemistry Figures*. *arXiv preprint*, 2025. **2507.20230**.
- Chen, Y., et al., *MolNexTR: A generalized deep learning model for molecular image recognition*. *Research Square*, 2024.
- Lowe, D.M., et al., *Chemical Name to Structure: OPSIN, an Open Source Solution*. *Journal of Chemical Information and Modeling*, 2011. **51**(3): p. 739–753.
- Kim, S., et al., *PubChem substance and compound databases*. *Nucleic acids research*, 2016. **44**(D1): p. D1202–D1213.
- NCI/CADD Chemical Identifier Resolver.
- Selector Group Chat — AutoGen.

16. Qian, Y., et al., *RxnScribe: A Sequence Generation Model for Reaction Diagram Parsing*. Journal of Chemical Information and Modeling, 2023. **63**(13): p. 4030–4041.
17. Landrum, G., *RDKit: Open-source cheminformatics*. 2024, RDKit.
18. *Introducing Claude Opus 4.7*.
19. *Claude Opus 4.6*.
20. *GPT-5.4 Model | OpenAI API*.
21. *GPT-5.5 Model | OpenAI API*.
22. *Gemini 3.1 Pro Preview | Gemini API*. Google AI for Developers.
23. *Gemini 3 Flash | Gemini Enterprise Agent Platform*. Google Cloud Documentation.
24. *Gemini 3.5 Flash*. Google DeepMind.
25. *Grok 4.3 | xAI Docs*.
26. *Qwen3.6-27B: Flagship-Level Coding in a 27B Dense Model*. Alibaba Cloud Community.